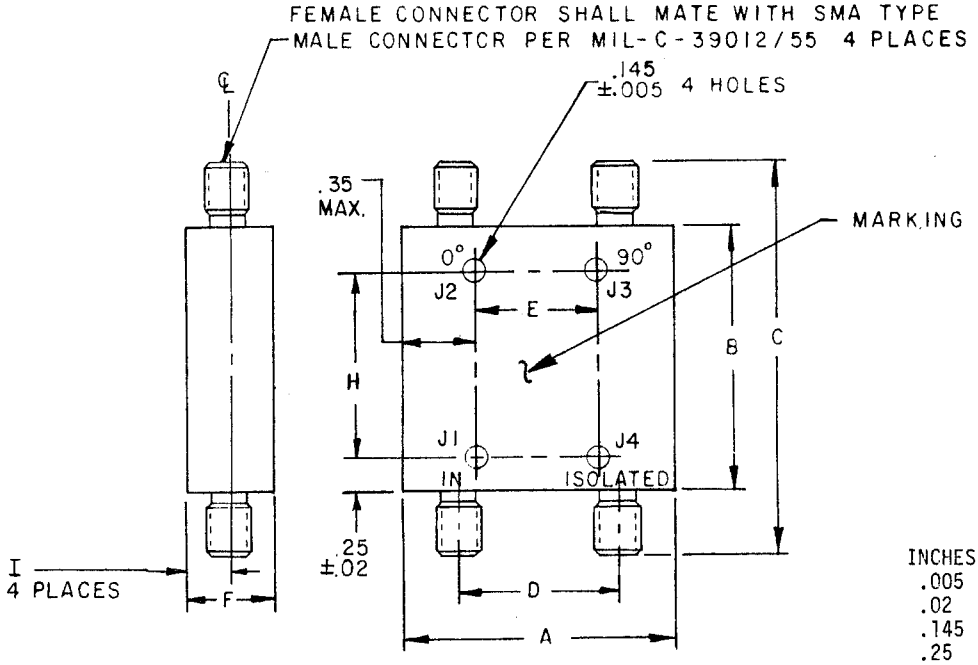
MILITARY SPECIFICATION SHEET

# POWER DIVIDERS, 4 PORT QUADRATURE, SMA CONNECTORS

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the power dividers described herein shall consist of this document and the latest issue of specification MIL-P-23971.




| INCHES | MM |
|---|---|
| .005 | .13 |
| .02 | .51 |
| .145 | 3.68 |
| .25 | 6.35 |
| .35 | 8.89 |

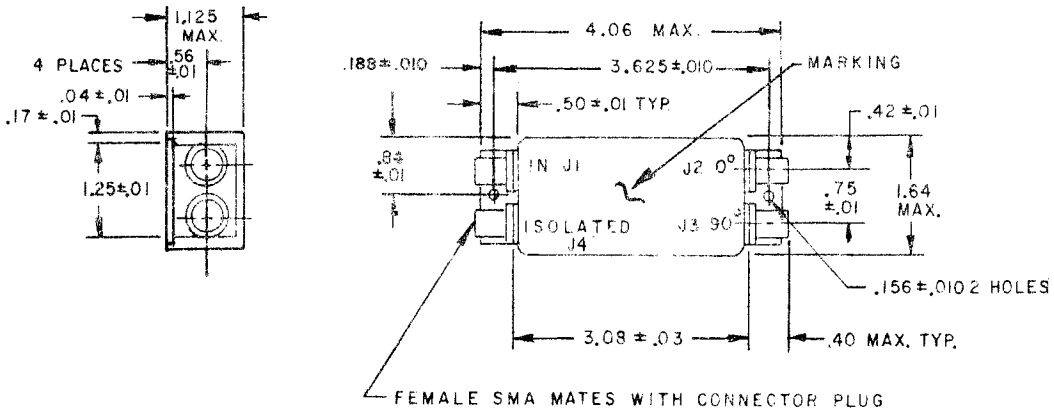
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Dimensions and configuration, dash numbers 01, 02, 03, and 04.



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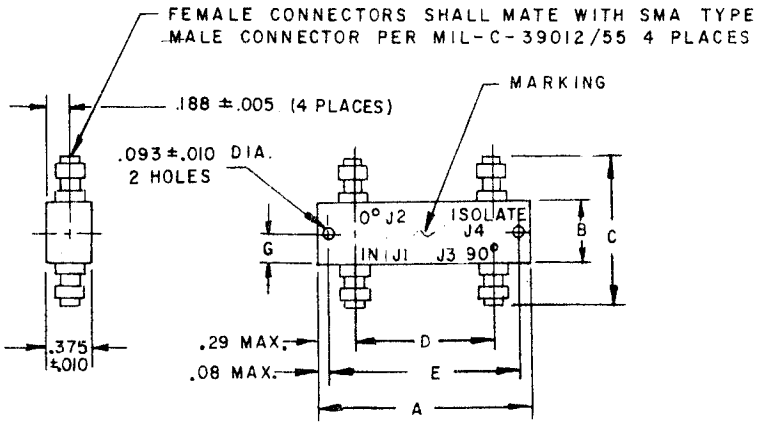
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| INCHES | MM |
|---|---|
| .01 | .3 |
| .03 | .8 |
| .04 | 1.0 |
| .010 | .25 |
| .156 | 3.96 |
| .17 | 4.4 |
| .188 | 4.77 |
| .40 | 10.2 |
| .42 | 10.7 |
| .50 | 12.7 |
| .56 | 14.2 |
| .75 | 19.0 |
| .84 | 21.3 |
| 1.125 | 28.57 |
| 1.25 | 31.8 |
| 1.64 | 41.6 |
| 3.08 | 78.2 |
| 3.625 | 92.07 |
| 4.06 | 103.1 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Dimensions and configuration, dash number 05.



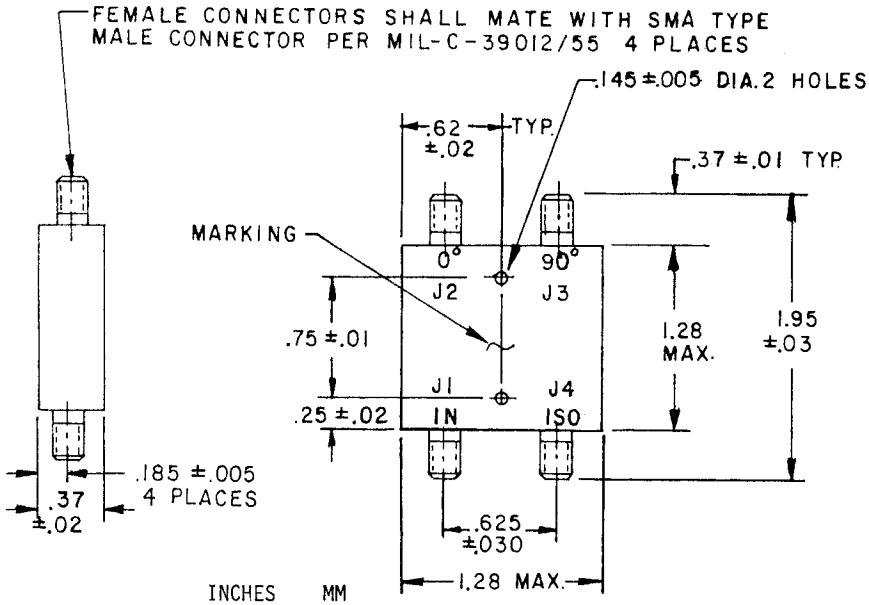

| INCHES | MM |
|---|---|
| .005 | .1 |
| .010 | .25 |
| .08 | 2.03 |
| .093 | 2.36 |
| .188 | 4.78 |
| .29 | 7.37 |
| .375 | 9.53 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm

FIGURE 3. Dimensions and configuration, dash numbers 06, 07, 08, 09, 11, and 12.



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| INCHES | MM |
|---|---|
| .005 | .1 |
| .01 | .3 |
| .02 | .5 |
| .03 | .8 |
| .030 | .76 |
| .145 | 3.68 |
| .185 | 4.70 |
| .25 | 6.35 |
| .37 | 9.40 |
| .62 | 15.7 |
| .625 | 15.9 |
| .75 | 19.0 |
| 1.28 | 32.5 |
| 1.95 | 49.5 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 4. Dimensions and configuration, dash number 10.



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Table I. Dimenisons

| M23971/2- | A | B | C | D | E | F | G | H | I | Figure |
|---|---|---|---|---|---|---|---|---|---|---|
| 01 | 4.000 ±.020 | 2.75 ±.02 | 3.500 ±.020 | 3.300 ±.020 | 1.300 ±.005 | .37 ±.02 | - | 2.25 ±.02 | .185 +.005 -.000 | 1 |
| 02 | 1.650 ±.020 | 1.62 ±.02 | 2.375 ±.020 | 1.000 ±.020 | - | .50 ±.02 | - - | 1.13 ±.02 | .250 +.005 -.000 | 1 |
| 03 | 1.250 ±.020 | 1.25 ±.02 | 2.000 ±.020 | .630 ±.020 | - - | .50 ±.02 | - - | .75 ±.02 | ↓ | 1 |
| 04 | 1.250 ±.020 | 1.25 ±.02 | 2.000 ±.020 | .630 ±.020 | - - | .50 ±.02 | - - | .75 ±.02 | ↓ | 1 |
| 06 | 3.320 | .530 | 1.160 ±.020 | 2.710 ±.020 | 3.130 ±.005 | - | .580 ±.020 | - | - | 3 |
| 07 | 2.582 | ↓ | ↓ | 1.972 ±.020 | 2.391 ±.005 | - | ↓ | - | - | 3 |
| 08 | 1.718 | ↓ | ↓ | 1.105 ±.020 | 1.525 ±.005 | - | ↓ | - | - | 3 |
| 09 | 1.160 | .530 | 1.160 ±.020 | .550 ±.020 | .969 ±.005 | - | ↓ | - | - | 3 |
| 11 | 1.108 | .655 | 1.285 ±.020 | .500 ±.020 | .922 ±.005 | - - | .642 ±.020 | - | - | 3 |
| 12 | 1.108 | .655 | 1.285 ±.020 | .500 ±.020 | .922 ±.005 | - - | ↓ | - - | - - | 3 |

| INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|
| .005 | .1 | .75 | 19.0 | 1.650 | 41.9 |
| .02 | .5 | .922 | 23.4 | 1.718 | 43.6 |
| .020 | .51 | .969 | 24.6 | 1.972 | 50.1 |
| .185 | 4.70 | 1.000 | 25.4 | 2.000 | 50.8 |
| .250 | 6.35 | 1.105 | 28.0 | 2.25 | 57.1 |
| .37 | 9.40 | 1.108 | 28.1 | 2.375 | 60.3 |
| .50 | 12.7 | 1.13 | 28.7 | 2.391 | 60.7 |
| .530 | 13.5 | 1.160 | 29.5 | 2.582 | 65.6 |
| .550 | 14.0 | 1.250 | 31.7 | 2.710 | 68.8 |
| .580 | 14.7 | 1.285 | 32.6 | 2.75 | 69.8 |
| .630 | 16.0 | 1.300 | 33.0 | 3.130 | 79.5 |
| .642 | 16.3 | 1.525 | 38.7 | 3.300 | 83.8 |
| .655 | 16.6 | 1.62 | 41.1 | 3.320 | 84.3 |
| | | | | 3.500 | 88.9 |
| | | | | 4.000 | 101.6 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

REQUIREMENTS:

Design and construction: See figures 1 through 4.

Case: O-ring sealed.

Electrical characteristics: See table II.

Weight: See table II.

Ambient temperature: See table II.

Environmental requirements: See table III.

Part number: M23971/2-(dash number from table II).



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TABLE II. Electrical performance characteristics and physical requirements.

| M23971/2- | Impedance (ohms) | Frequency range GHz | Average coupling (dB) | VSWR max | Insertion loss max (dB) | Isolation min (dB) | Phase balance max 90° | Amplitude balance (dB) max | Power level avg (W) | Power level pk (W) | Weight pounds (grams) | Ambient temperature Operating | Ambient temperature Storage | Figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 01 | 50 | 0.25-1.0 | 3. +.2 -.0 | 1.3:1 | 0.5 | 20 | ±1.5 | ±0.75 | 100 | 5K | .41 (187) | -54° to +71°C | -62° to +85°C | 1 |
| 02 | 50 | 1.0-2.0 | 3. +.2 -.0 | 1.3:1 | 0.25 | 23 | ±1.5 | ±0.5 | 100 | 5K | .10 (45.4) | ↓ | ↓ | 1 |
| 03 | 50 | 2.0-4.0 | 3. +.2 -.0 | 1.3:1 | 0.25 | 20 | ±1.5 | ±0.5 | 100 | 5K | .10 (45.4) | ↓ | ↓ | 1 |
| 04 | 50 | 4.0-8.0 | 3. +.2 -.0 | 1.35:1 | 0.35 | 18 | ±1.5 | ±0.5 | 100 | 5K | .16 (70.9) | -54° to +71°C | -62° to +85°C | 1 |
| 05 | 50 | .02-.200 | 3. +.5 -.0 | 1.3:1 | 1.3 | 20 | ±5 | ±1.0 | 1 | - | - | -40° to +54°C | -40° to +54°C | 2 |
| 06 | 50 | .5-1.0 | 3. +.2 -.0 | 1.25:1 | 0.25 | 18 | ±1.5 | ±0.5 | 50 | 250 | .14 (63.5) | -54° to +110°C | -54° to +110°C | 3 |
| 07 | 50 | 1.0-2.0 | 3. +.2 -.0 | 1.25:1 | 0.25 | 18 | ±1.5 | ±0.5 | 50 | 250 | .11 (49.9) | -54° to +110°C | -54° to +110°C | 3 |
| 08 | 50 | 2.0-4.0 | 3. +.2 -.0 | 1.25:1 | 0.25 | 18 | ±2.0 | ±0.5 | 50 | 250 | .07 (31.8) | -54° to +110°C | -54° to +110°C | 3 |
| 09 | 50 | 4.0-8.0 | 3. +.2 -.0 | 1.25:1 | 0.35 | 18 | ±2.0 | ±0.5 | 50 | 250 | .05 (22.7) | ↓ | ↓ | 3 |
| 10 | 50 | 7.0-11.0 | 3. +.5 -.0 | 1.45:1 | 0.50 | 17 | ±2.0 | ±0.5 | 100 | 5K | .11 (49.9) | ↓ | ↓ | 4 |
| 11 | 50 | 8.0-12.4 | 3. +.2 -.0 | 1.30:1 | 0.40 | 15 | ±3.0 | ±0.5 | 50 | 250 | .06 (27.2) | ↓ | ↓ | 3 |
| 12 | 50 | 12.4-18.0 | 3. +.2 -.0 | 1.35:1 | 0.50 | 15 | ±5.0 | ±0.5 | 50 | 250 | .06 (27.2) | -54° to +110°C | -54° to +110°C | 3 |





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TABLE III. Environmental tests: In accordance with MIL-P-28971 except:

| M23971/2– | Barometric pressure | Thermal shock | Salt spray | Explosion | Vibration | Pressurization |
|---|---|---|---|---|---|---|
| 01, 02, 03, 04 | test condition A | 3/ | 3/ | applicable | 3/ | 3/ |
| 06, 07, 08, 09, 10, 11, 12 | 3/ | 3/ | test condition A | 3/ | test condition B | 1/ 2/ |

1/ Pressurization: Power dividers shall be subjected to an internal gas pressure of 30 pounds per square inch for at least 20 seconds while immersed in water at approximately 20°C. During the test, there shall be no leakage, as detected by the continuous formation of escaping air bubbles. After the test, VSWR, insertion loss, isolation, phase balance, and amplitude balance tests shall be performed in accordance with 4.7.4, 4.7.5, 4.7.6, 4.7.7, and 4.7.8 of Mil-P-23971 respectively.
2/ Pressurization shall replace the barometric test.
3/ Not applicable.

TABLE IV. Supersession data.

| Part number M23971/2– | Supersedes M15370/8 |
|---|---|
| 06 | 001 |
| 07 | 002 |
| 08 | 003 |
| 09 | 004 |
| 11 | 005 |
| 12 | 006 |
| 10 | 007 |

NOTE: Power dividers covered by this specification sheet are substituted for the following manufacturer's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

| Dash number | Manufacturer's code | Manufacturer's part number | FSN |
|---|---|---|---|
| 01 | 32562<br>31597 | USCD-10115-4<br>10023-3 | 5985-00-459-0769 |
| 02 | 32562<br>31597 | USCD-10115-1<br>10015-3 | 5985-00-442-4718<br>5985-00-459-2039 |
| 03 | 32562<br>31597 | USCD-10115-2<br>10016-3 | 5985-00-442-4730<br>5985-00-428-0710 |
| 04 | 32562<br>31597 | USCD-10116-3<br>10017-3 | 5985-00-428-0682<br>5985-00-428-0682 |



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| Dash number | Manufacturer's code | Manufacturer's part number | FSN |
|---|---|---|---|
| 05 | 12457 | QHM-7-122/20079 | — |
| 06 | 07397<br>16179 | 20-80546-4<br>20152-3 | ↓ |
| 07 | 07397<br>16179<br>72152 | 20-80546-5<br>20153-3<br>900175-1 | 5985-00-927-7535 |
| 08 | 07397<br>16179 | 20-80546-6<br>20154-3 | ↓ |
| 09 | 07397<br>16179<br>94990 | 20-80546-7<br>20155-3<br>58-30723C01 | —<br>5985-00-498-5566<br>↓ |
| 10 | 05869<br>31597 | 721701-1<br>1B0018-3 | ↓ |
| 11 | 07397<br>16179 | 20-808546-8<br>20156-3 | ↓ |
| 12 | 16179<br>06424<br>81413 | 20157-3<br>10164-1<br>16181-003 | 5985-00-108-4989<br>↓<br>— |

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army – ER
Navy – EC
Air Force – 85

Review activities:
Army –
Navy –
Air Force –
DLA – ES

User activities:
Army –
Navy – AS, OS, MC
Air Force –

Preparing activity:
Navy – EC

Agent:
DLA – ES

(Project 5985-0890)

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